ALU with Built-In Self-Test (BIST), Power Gating, and Dynamic Voltage & Frequency Scaling (DVFS) Optimization

Course: CMPE 630 – Digital IC Design
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Overall Block Diagram

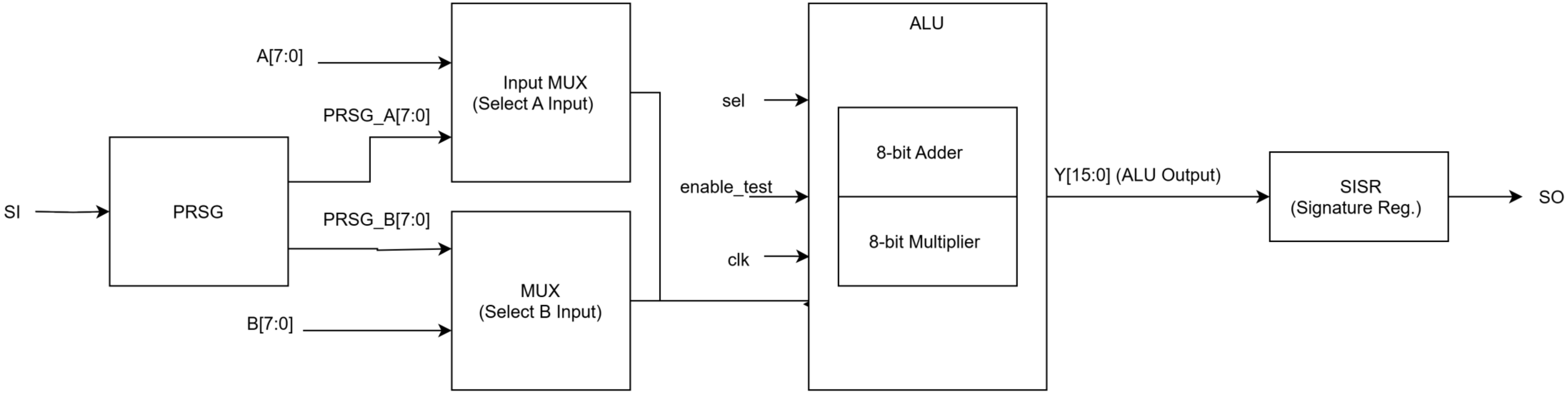

Figure 1: Block diagram of the ALU with Built-In Self-Test (BIST) integration. The Pseudo-Random Sequence Generator (PRSG) supplies randomized test inputs through 2-to-1 multiplexers, allowing selection between normal operation and test mode. The ALU performs either addition or multiplication based on the control signal sel. The result is compressed by the Single Input Signature Register (SISR) into a compact signature, which is scanned out via SO for verification.

Block Diagram Legend

Symbol/Label	Description
A[7:0]	8-bit external input A to ALU
B[7:0]	8-bit external input B to ALU
PRSG	Pseudo-Random Sequence Generator; generates 8-bit test values for A and B
SI	Serial Input (Scan In) to load seed into PRSG during test mode
MUX (A/B)	2-to-1 MUXes select between normal inputs and PRSG outputs
enable_test	Control signal: 1 = use PRSG inputs and enable SISR
sel	Control signal: 0 = Adder, 1 = Multiplier in ALU
clk	Clock signal for synchronous elements (PRSG, SISR, ALU if sequential)
ALU	Arithmetic Logic Unit – executes either A + B or A × B
8-bit Adder	Ripple-carry or carry-lookahead adder using standard cells
8-bit Multiplier	Array or shift-add style multiplier using logic gates
Y[15:0]	16-bit output from ALU (for mult and add)
SISR	Single Input Signature Register – compresses output Y to a small signature
SO	Scan Out – the final compressed signature for comparison to golden sig

Major Component – ALU (Adder + Multiplier)

- Adder: 8-bit ripple-carry adder from Lab 8 using standard cells (AND, XOR, INV)
- Multiplier: New 8-bit unsigned array multiplier built using AND and adder stages
- MUX: Chooses between Adder and Multiplier output using sel bit
- Output: Y[15:0]

Major Component – PRSG + MUX

- PRSG (Pseudo-Random Sequence Generator):
  - 9-bit Galois LFSR
  - Used to generate randomized test inputs

- Input MUXes:
  - Two 2-to-1 MUXes select between external A/B and PRSG outputs
  - Controlled by enable_test signal

Major Component – SISR

- SISR (Single Input Signature Register):
  - 6-bit register with feedback XOR logic
  - Shifts in 1-bit responses from ALU output (e.g., parity or bitwise XOR of Y)
  - Final value is compared to golden signature for pass/fail validation

Optimization – Power Gating and DVFS

- Goal: Save power and improve energy efficiency using multiple strategies
- Power Gating Method:
  - Use sleep transistors to cut VDD to inactive blocks (e.g., Multiplier, PRSG, SISR)
  - Control logic disables power to idle blocks to reduce leakage
- DVFS Method:
  - Adjust operating voltage and frequency dynamically based on workload
  - Lower frequency/voltage for test mode; higher for performance-critical operations
- Why Combine Power Gating and DVFS:
  - Power gating reduces static/leakage power during idle time
  - DVFS reduces dynamic power based on activity level
  - Together, these techniques allow fine-grained control over power and performance tradeoffs, helping to meet timing and energy constraints across different operating modes
  - This combination aligns with ongoing research into adaptive low-power design, and provides a foundation for exploring DVFS schemes in dissertation-level investigations

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THANK YOU